REPRESENTABLE FLATNESS IMPLIES FLATNESS FOR MORPHISMS OF FORMAL ALGEBRAIC SPACES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the source proves the lemma by a four-implication flatness chain on affine charts and closes with the limit flatness lemma; the argument is complete

Editorial note. The label lemma-representable-flat names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- morphisms of locally Noetherian formal algebraic spaces representable by algebraic spaces and flat in the transformation sense of the bootstrap chapter
- flatness of morphisms of formal algebraic spaces defined via the taut maps of the WAdm^Noeth category on affine etale charts
- etale morphisms are flat and flatness composes
- the etale-flat up-down lemma for flatness after etale base change
- flatness of a limit of ring maps from flatness of the reductions

Source reference. Original definition 03XZ: sources/stacks-chow/tags.txt, source lines 5166–5166. Prerequisite pointer invoked at proof line(s) 3141; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 00HB: sources/stacks-restricted/pinned-original-source.tex, source lines 3023–3023. Prerequisite pointer invoked at proof line(s) 3142; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 05B9: sources/stacks-chow/tags.txt, source lines 6877–6877. Prerequisite pointer invoked at proof line(s) 3194; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0GBS: sources/stacks-chow/tags.txt, source lines 20382–20382. Prerequisite pointer invoked at proof line(s) 3165; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

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Source reference. Original lemma 0ANK: sources/stacks-chow/tags.txt, source lines 13443–13443. Prerequisite pointer invoked at proof line(s) 3163; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0912: sources/stacks-chow/tags.txt, source lines 11430–11430. Prerequisite pointer invoked at proof line(s) 3205; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphisms of locally Noetherian formal algebraic spaces over S. If f is representable by algebraic spaces and flat in the sense of Bootstrap, Definition 03XZ, then f is flat in the sense of Definition 00HB.*

Proof. This is a sanity check whose proof should be trivial but isn't quite. We urge the reader to skip the proof. Assume f is representable by algebraic spaces and flat in the sense of Bootstrap, Definition 03XZ. Consider a commutative diagram

$$\begin{array}{ccc} U & \longrightarrow & V \\ \downarrow & & \downarrow \\ X & \longrightarrow & Y \end{array}$$

with U and V affine formal algebraic spaces, $U \to X$ and $V \to Y$ representable by algebraic spaces and étale. Then the morphism $U \to V$ corresponds to a taut map $B \to A$ of $WAdm^{Noeth}$ by Formal Spaces, Lemma 0ANK. Observe that this means $B \to A$ is adic (Formal Spaces, Lemma 0GBS) and in particular for any ideal of definition $J \subset B$ the topology on A is the J-adic topology and the diagrams

$$\begin{array}{ccc} \mathrm{Spec}(A/J^nA) & \longrightarrow & \mathrm{Spec}(B/J^n) \\ \downarrow & & \downarrow \\ U & \longrightarrow & V \end{array}$$

are cartesian.

Let $T \to V$ is a morphism where T is a scheme. Then

$$\begin{aligned} X \times_Y T \to T \text{ is flat} &\Rightarrow U \times_Y T \to T \text{ is flat} \\ &\Rightarrow U \times_V V \times_Y T \to T \text{ is flat} \\ &\Rightarrow U \times_V V \times_Y T \to V \times_Y T \text{ is flat} \\ &\Rightarrow U \times_V T \to T \text{ is flat} \end{aligned}$$

The first statement is the assumption on f. The first implication because $U \to X$ is étale and hence flat and compositions of flat morphisms of algebraic spaces are flat. The second impliciation because $U \times_Y T = U \times_V V \times_Y T$. The third implication by More on Flatness, Lemma 05B9. The fourth implication because we can pullback by the morphism $T \to V \times_Y T$. We conclude that $U \to V$ is flat in the sense of Bootstrap, Definition 03XZ. In terms of the continuous ring map $B \to A$ this means the ring maps $B/J^n \to A/J^nA$ are flat (see diagram above).

Finally, we can conclude that $B \to A$ is flat for example by More on Algebra, Lemma 0912. □